

Customer Shopping Behavior Analysis

1. Business Problem Statement

A retail company aims to understand customer shopping behavior to improve sales, customer satisfaction, and long-term loyalty. The company has observed changes in purchasing patterns across demographics, product categories, and shopping channels (online vs offline). The objective is to identify key factors influencing purchasing decisions such as discounts, reviews, seasons, and payment preferences.

2. Project Objective

Analyze customer shopping data to uncover trends, segment customers, and generate actionable insights that support better marketing strategies, product optimization, and customer engagement.

3. Dataset Overview

The dataset consists of 3,900 transactions with 18 features including customer demographics, purchase details, and shopping behavior such as discount usage, review ratings, and purchase frequency.

4. Methodology

Data Preparation (Python): Cleaned missing values, standardized columns, and performed feature engineering.

Data Analysis (SQL): Executed queries to identify revenue trends, customer segments, and purchase behavior.

Visualization (Power BI): Built dashboards to present insights for decision-making.

5. Key Insights

- High-spending customers are not always discount-dependent.
- Subscription users contribute higher overall revenue.
- Certain product categories perform better in specific seasons.
- Loyal customers show higher repeat purchase behavior.

6. Business Recommendations

- Enhance loyalty programs to retain repeat customers.
- Optimize discount strategies to maintain profit margins.
- Focus marketing on high-performing customer segments.
- Promote top-rated and high-demand products.
- Encourage subscriptions through exclusive benefits.

7. Conclusion

This project demonstrates how data-driven insights can help businesses better understand customer behavior, improve engagement, and optimize overall strategy for sustained growth.